

Homework (Habanero)

Q1. Factor the trinomial $x^2 + 5x + 6$.

- (A) $x(x + 3) + 2(x + 3)$
- (B) $(x + 2)(x + 3)$
- (C) $(x + 1)(x + 6)$
- (D) $x(x + 2) + 3(x + 2)$
- (E) $(x + 4)(x + 2)$

Q2. The population of a certain bacteria colony grows according to the equation $P(t) = t^2 + 4t + 3$, where t is time in hours. Factor $P(t)$.

- (A) $(t + 1)(t + 3)$
- (B) $(t + 2)(t + 2)$
- (C) $(t + 4)(t + 1)$
- (D) $(t + 3)(t + 2)$
- (E) $(t + 2)(t + 1)$

Q3. A pharmaceutical company produces a drug with a concentration given by $C(x) = x^2 - 7x + 12$. Factor $C(x)$.

- (A) $(x - 3)(x - 4)$
- (B) $(x - 2)(x - 5)$
- (C) $(x + 3)(x - 4)$
- (D) $(x - 1)(x - 6)$
- (E) $(x + 2)(x - 6)$

Q4. The rate at which a patient's body absorbs a certain medication is given by $R(t) = t^2 - 9t + 20$. Factor $R(t)$.

- (A) $(t - 4)(t - 5)$
- (B) $(t - 2)(t - 10)$
- (C) $(t + 4)(t - 5)$
- (D) $(t - 1)(t - 8)$

(E) $(t + 1)(t - 9)$

Q5. Factor the trinomial $x^2 - 2x - 15$.

- (A) $(x + 3)(x - 5)$
- (B) $(x - 3)(x + 5)$
- (C) $(x + 2)(x - 7)$
- (D) $(x - 2)(x + 8)$
- (E) $(x + 1)(x - 9)$

Q6. The spread of a virus can be modeled by $V(t) = t^2 + 2t - 6$. Factor $V(t)$.

- (A) $(t + 3)(t - 2)$
- (B) $(t - 3)(t + 2)$
- (C) $(t + 2)(t + 3)$
- (D) $(t + 1)(t - 6)$
- (E) $(t - 1)(t + 6)$

Q7. A patient's temperature $T(x)$, in degrees Celsius, x hours after taking medication is $T(x) = x^2 - x - 12$. Factor $T(x)$.

- (A) $(x - 4)(x + 3)$
- (B) $(x - 2)(x + 6)$
- (C) $(x - 3)(x + 4)$
- (D) $(x + 2)(x - 6)$
- (E) $(x + 1)(x - 12)$

Q8. Factor $x^2 + 10x + 24$.

- (A) $(x + 4)(x + 6)$
- (B) $(x + 3)(x + 8)$
- (C) $(x + 2)(x + 12)$
- (D) $(x + 1)(x + 24)$
- (E) $(x + 5)(x + 5)$

Open Ended Questions (FRQ)

Q9. The growth rate of a certain tumor is given by the equation $G(t) = t^2 - 5t - 14$. Factor $G(t)$ and explain the significance of factoring in understanding tumor growth.

Q10. The concentration of a solution $C(x)$ after x minutes of a chemical reaction is $C(x) = x^2 + 8x + 12$. Factor $C(x)$ and describe how factoring can help in predicting the behavior of chemical reactions.

Chef's Tips

- Read the questions carefully and allocate time wisely.
- For multiple-choice questions, choose the best answer.
- For free-response questions, show all work and explain your reasoning.
- Use the space provided for your answers and do not write on the margins.